

Module 5: answer keys

CFDE lung transcriptomics course

Instructor document. Do not distribute to learners.

Knowledge check: answer key

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Ten single-answer items, used unchanged as pre-test and post-test. One point each, ten points total. Every item has one defensible answer; the distractors are misconceptions this course corrects rather than invented alternatives.

#	LO	Answer	Why, and the misconception it corrects
1	LO1	C	Size and checksum establish file identity, nothing more. They say the bytes match the ones the analysis was written against. B is the common conflation of provenance with quality: a verified file can still be unfit for the question. D confuses identity with equivalence, which is exactly what the primary versus processed distinction denies.
2	LO1	B	The discovery layer tells you which resources hold matching data; retrieval still happens at each portal. A is the persistent expectation of a single harmonized download, which the ecosystem does not provide.
3	LO3	D	Harmony adjusts the PCA embedding and stores it as <code>X_pca_harmony</code> ; the counts are untouched. Mixing metrics and the donor-colored UMAP are statements about that embedding, which is why neighbors and clustering change and differential expression does not run on corrected values. A is the consequential version of the error, since a learner who believes it will treat donor as already handled in the DE model. B and C extend the

			correction to objects it never writes.
4	LO2	B	A threshold is justified for the assay or shown not to drive the result. A is the habit the objective targets: inheriting a value because it is common. C is wrong on the biology, since nuclear preparations still carry mitochondrial reads. D tunes QC until donors agree, which manufactures the comparability the analysis is supposed to test.
5	LO3	A	Frequency is composition. Reading it as per-cell expression (B) is the most common single-cell error, and separating the two is the stated objective of Module 2.
6	LO3	C	The background sets what counts as surprising. Using the whole library rather than the measured gene universe inflates significance, and the effect is worst on short input lists.
7	LO4	A	Variance explained in one view and near-zero in the other means the factors are modality-private: the method found no shared axis. B mistakes a clean separation for successful integration.
8	LO5	D	Native units are not comparable across platforms and species; a within-resource contrast shares a design, so the contrast is what travels. C invents a property that pathway scores do not have.
9	LO5	A	A coefficient over every shared pathway averages the passing set together with the pathways that failed, so it cannot say whether the passers agree beyond chance. That is why Module 4 demotes the global correlation to supporting context and makes the permutation screen the headline. C is the intuition the item targets: more pathways

			raise precision on the wrong quantity.
10	LO5	C	Report at the granularity the analysis produced. A and B are the same error in two forms: co-occurrence in one direction is not evidence of shared regulation.

Answer string: C B D B A C A D A C

Coverage and item quality, measured

Recorded so a later editor can confirm an edit has not reintroduced a test-wise cue.

- **Learning objective coverage:** LO1 two items, LO2 one, LO3 three, LO4 one, LO5 three. LO5 carries three because it spans Modules 4 and 5 and contains two distinct clauses: comparing at the level of contrasts rather than values, and stating what the analysis does not show.
- **Answer key distribution:** A three times, B twice, C three times, D twice.
- The correct answer is the longest option in **0 of 10** items, so length carries no signal about the key. Check this again after any edit to an option; it is the cue most easily reintroduced.
- Items 1 to 3 depend on the Module 1 screencast and the "Getting the data" section of the Module 1 document. Do not remove either without re-basing those items.
- **The learner instrument carries no LO tags.** The course learning objectives are numbered only in the instructional design record, which is not part of the learner package, so the tags were meaningless on the page. The mapping is kept here, in the column above, where it is used.

If you need a harder instrument

Items 5 to 10 were previously paired with a written justification worth a second point. That version scored out of 16 and does not fit a ten-minute window. If the assessment is given untimed, or as a take-home, asking for one to three sentences on items 5, 7 and 10 recovers most of the discrimination at modest cost: those three separate a learner who recognizes the correct answer from one who can say why composition is not expression, why modality-private factors mean no shared axis, and why co-occurrence in one direction does not license naming a mechanism.

Capstone exercise: model answer and rubric

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The capstone asks for a brief abstract applying the course approach to a new question, with the learner locating the data themselves. There is no single correct answer, and no target length. The rubric rewards a design whose limits are visible.

What a strong response shows

Question. Names a tissue or system and a contrast, not a topic. "Does alveolar macrophage phenotype differ between smokers and non-smokers in human lung?" is answerable. "Study lung inflammation" is not.

Data and discovery. This carries the most weight, because locating the data is the part that most resembles real work. A strong response shows the candidate resources were actually inspected: what the resource contains, whether its metadata records the variable in question, and what could be obtained without special access. Naming a resource without saying what it contributes is the most common shortfall. Honest identification of external data where the Common Fund does not cover the question earns full marks; Module 4 does exactly this for fibrotic human lung.

Design. Case and reference groups for each resource. Credit responses that state a reference level explicitly, since Module 4 showed that alphabetical ordering would have inverted three of four contrasts.

What travels. Names both the comparable quantity, such as log fold change or pathway enrichment score, and the quantity that must not be compared across platforms, species or units, with a reason rather than an assertion.

Limits. A real limitation tied to this design, with the accompanying disclosure. Group sizes, annotation or ortholog loss, a null model for apparent agreement, or a calibration check on a suspect contrast are all creditable.

Common weaknesses

- A question that describes rather than contrasts.
- Resources named but not characterised, suggesting the discovery step was skipped.
- Proposing to merge counts or expression values across resources, which the course argues against throughout.
- Limitations stated generically ("more samples would help") rather than tied to the design.
- Claiming a resource holds data it does not. Responses should be checkable against the portal.

Scoring

Question 20, Data and discovery 30, Design 20, What travels 15, Limits 15.

Full marks require a design whose limits are visible in the design itself, not a novel study. A learner who discovers mid-exercise that their preferred resource cannot answer their question, and says so, has demonstrated the intended skill.